

MINE E 630: Underground Mining and Bulk Materials Handling
Assignment #1

Due Date: January 22, 2026 (until 11: 59 pm)

Method of Submission: Please provide detailed solutions and submit a combined pdf file to the eClass assignment before the specified due time.

1) The portal for an exploration decline will be at 10,500 ft N; 6,400 ft E; and an elevation of 4,260 ft. The bottom of the decline will be at 9,600 ft N; 7,000 ft E; and an elevation of 4,010 ft. What is the length (feet) of the decline? (20 points)

2) An ore deposit has a uniform geometry and the following characteristics.

- Strike distance: 400 m
- Dip distance: 500 m
- Dip of ore body: 85° from horizontal
- Top of ore body thickness: 10 m
- Bottom of ore body thickness: 20 m
- Density: 2.45 tonnes/m³

How many tonnes does this ore deposit have? (20 points)

3) A tabular vein with outcrops has the strike 70°NW dipping 50° . Some dimensions are listed as follows (20 points),

- (a) Width: 100 m (along with the direction of the strike)
- (b) Depth (from the bottom): 200 m
- (c) Thickness: 20 m

Requirements:

1. Please draw a freehand isometric sketch for this vein. (not to scale)
2. Annotate the angle in degrees to show the north, strike, and dip.
3. Annotate all necessary dimensions of the vein (length, width, depth, and thickness) on the sketch.
4. Mark the location of the hanging wall and footwall.
5. Draw a schematic drift and a crosscut in the vein.

4) Selection of Underground Mining Methods (40 points)

The conditions of the hangwall, orebody, and footwall are given by a hard rock mine.

- Shape: Tabular
- Average Thickness: 88 m
- Dipping angle: 85°
- Grade Distribution: Erratic
- Average deposit depth (to ground surface): 1200 m
- Average specific gravity of overburden: 2.7
- Average uniaxial compressive strength (UCS):
 - Hanging wall: 91 MPa
 - Ore: 103 MPa
 - Footwall: 125 MPa
- Average Rock Quality Designation (sampled by mine geologists):
 - Hanging wall: 27%
 - Ore: 50%
 - Footwall: 72%
- Fracture strength (sampled by mine geologists): Moderate for both host rock and ore

As a mine engineer,

- (1) Please hand calculate the rock substance strength of the hanging wall, ore, and footwall. Write down the process in this assignment. (20 points)
- (2) Please decide the best two underground mining methods using the UofA Macdonald Mining Method Selection Tool (<https://sites.ualberta.ca/%7Ewvliu/Mining%20Method%20Selection.html>). (20 points)
 - a. What are the scores of the two underground methods?
 - b. Please make a snapshot of the whole screen of the tool.